Power of Two (Data Size Units)

When working with **system design and distributed systems**, understanding **binary-based data units (power of 2)** is very important for accurate storage and memory estimation.

1. Why Power of 2 Matters

Computers use **binary (base-2)** internally, so storage is naturally measured in powers of 2:

None

```
1 KB = 2^10 bytes = 1024 bytes

1 MB = 2^20 bytes

1 GB = 2^30 bytes

1 TB = 2^40 bytes
```

This helps avoid estimation errors in large-scale systems.

2. Basic Unit: Bit and Byte

None

```
1 bit  = 0 or 1

1 byte = 8 bits
```

A byte typically stores:

None
```
1 ASCII character
```

Example:

None
```
'A' = 1 byte

'Z' = 1 byte

'0' = 1 byte
```

3. Data Units Table (Power of 2)

Storage Scale

None
```

1 KB = 2¹⁰ = 1,024 bytes

1 MB = 2¹⁰ KB = 1,024 KB

1 GB = 2¹⁰ MB = 1,024 MB

1 TB = 2¹⁰ GB = 1,024 GB

1 PB = 2¹⁰ TB = 1,024 TB

1 EB = 2¹⁰ PB = 1,024 PB
```

4. Quick Reference (Interview View)

Unit	Bytes (Approx Power of 2)
KB	$10^3 \approx 2^{10}$ = 1,024 bytes
MB	2^{20} bytes
GB	2^{30} bytes
TB	2^{40} bytes
PB	2^{50} bytes

5. Why 1024 Instead of 1000?

Because:

```None
Computers use binary addressing

Memory is divided in powers of 2
```

So:

```None
2¹⁰ = 1024 ≈ 10³
```

For estimation, we often approximate:

```None
1024 ≈ 1000 (for simplicity)
```

6. Example in System Design

Example: Storage estimation

Suppose:

```None
1 user profile = 2 KB

1 million users
```

Total storage:

```None
2 KB × 1,000,000 = 2,000,000 KB

≈ 2,000 MB

≈ 2 GB
```

7. Why This Is Important

In interviews, this helps you:

```None
Estimate DB size

Estimate cache size

Estimate bandwidth

Choose storage systems

Understand scaling limits
```

8. Common Mistake

❌ Wrong assumption:

```None
1 KB = 1000 bytes always
```

✔ Correct understanding:

```None
1 KB = 1024 bytes (binary systems)
```

But in rough estimation:

```None
We may approximate 1024 ≈ 1000
```

9. One-Line Summary

Power of two units define storage sizes using binary multiples (1024-based scaling), which is essential for accurate memory, storage, and bandwidth estimation in system design.